Sample Size & Effect Size Calculation

Empirical Bayes Contextual Bandit

Claude

May 10, 2026

Contents

1 Sample Size Formula for Longitudinal Bandit Studies

1.1 Model Setup

The reward model is

$$R_{i,t}(a) = \theta_{i,a,0} + \theta_{i,a,1}\, c_{i,t} + \varepsilon_{i,t,a}, \qquad \varepsilon_{i,t,a} \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2),$$

where $i \in [n]$ indexes participants, $t \in [T]$ indexes time points, $a \in \{0, 1\}$ is the action, and $c_{i,t} \in \{0, 1\}$ is the binary context variable with $P(c_{i,t} = 1) = p$ for all i, t. Each participant's parameters are drawn from a population hyperdistribution:

$$\boldsymbol{\theta}_{i,a} \sim \mathcal{N}(\boldsymbol{\mu}_a,\, \boldsymbol{\Sigma}_a), \qquad \boldsymbol{\mu}_a = \begin{pmatrix} \mu_{a,0} \\ \mu_{a,1} \end{pmatrix}, \quad \boldsymbol{\Sigma}_a = \begin{pmatrix} \tau^2_{a,00} & \tau_{a,01} \\ \tau_{a,10} & \tau^2_{a,11} \end{pmatrix}.$$

Here $\mu_{a,0}$ is the population mean baseline reward under arm a, and $\mu_{a,1}$ is the population mean context slope under arm a. The diagonal entries $\tau^2_{a,00}$ and $\tau^2_{a,11}$ are the between-person variances of the baseline and context slope respectively, and $\tau_{a,01}$ is their covariance within arm a.

1.2 Treatment Effects by Context

The treatment effect at context $c \in \{0, 1\}$ is

$$\delta_c = \big(\mu_{1,0} + \mu_{1,1} \cdot c\big) - \big(\mu_{0,0} + \mu_{0,1} \cdot c\big).$$

Explicitly:

$$\begin{aligned} \delta_0 &= \mu_{1,0} - \mu_{0,0} && \text{(treatment effect at } c = 0) \\ \delta_1 &= (\mu_{1,0} + \mu_{1,1}) - (\mu_{0,0} + \mu_{0,1}) && \text{(treatment effect at } c = 1) \end{aligned}$$

If $\mu_{1,1} \neq \mu_{0,1}$, the treatment effect differs by context (interaction effect). The **marginal treatment effect**, averaging over the context distribution, is

$$\delta = (1 - p)\, \delta_0 + p\, \delta_1 = (\mu_{1,0} - \mu_{0,0}) + p\, (\mu_{1,1} - \mu_{0,1}).$$

The standardized marginal effect size is $d = \delta/\sigma$.

1.3 Variance of the Estimated Treatment Effect

Let T_a denote the number of times arm a is pulled per participant ($T_0 + T_1 = T$). Person i's average reward under arm a, averaging over the T_a time points at which arm a was pulled, is

$$\bar{R}_{i,a} = \theta_{i,a,0} + \theta_{i,a,1}\, \bar{c}_{i,a} + \bar{\varepsilon}_{i,a},$$

where $\bar{c}_{i,a}$ is the empirical proportion of $c_{i,t} = 1$ among the pulls of arm a, which concentrates around p for large T_a. Person i's estimated treatment effect is

$$\hat{\delta}_i = \bar{R}_{i,1} - \bar{R}_{i,0} \approx (\theta_{i,1,0} - \theta_{i,0,0}) + (\theta_{i,1,1} - \theta_{i,0,1})\, p + \bar{\varepsilon}_{i,1} - \bar{\varepsilon}_{i,0}.$$

This is an estimator of the marginal treatment effect δ. Its variance has two components:

$$\operatorname{Var}(\hat{\delta}_i) = \underbrace{(\tau^2_{1,00} + \tau^2_{0,00}) + p^2(\tau^2_{1,11} + \tau^2_{0,11}) + 2p(\tau_{1,01} + \tau_{0,01})}_{\text{between-person heterogeneity in treatment effect}} + \underbrace{\sigma^2\left(\frac{1}{T_1} + \frac{1}{T_0}\right)}_{\text{within-person estimation noise}}.$$

The three terms in the between-person component are:

- $\tau^2_{1,00} + \tau^2_{0,00}$: variation across people in their baseline responses to each arm (relevant at any p).

- $p^2(\tau_{1,11}^2+\tau_{0,11}^2)$: variation across people in how much their context slope differs between arms, weighted by how often $c=1$ occurs. Disappears at $p=0$.
- $2p(\tau_{1,01}+\tau_{0,01})$: contribution from the within-arm covariance between baseline and context slope. Disappears at $p=0$.

The population-level estimator $\hat{\delta}=\frac{1}{n}\sum_i \hat{\delta}_i$ has variance

$$\mathrm{Var}(\hat{\delta})=\frac{1}{n}\left[(\tau_{1,00}^2+\tau_{0,00}^2)+p^2(\tau_{1,11}^2+\tau_{0,11}^2)+2p(\tau_{1,01}+\tau_{0,01})+\sigma^2\left(\frac{1}{T_1}+\frac{1}{T_0}\right)\right].$$

1.4 Sample Size Formula

Step 1: the rejection rule. We test $H_0:\delta=0$ vs. $H_1:\delta\neq 0$. By the central limit theorem, $\hat{\delta}$ is approximately normal for large n. Standardizing, the test statistic

$$Z=\frac{\hat{\delta}}{\sqrt{\mathrm{Var}(\hat{\delta})}}$$

is approximately $\mathcal{N}(0,1)$ under H_0 and $\mathcal{N}(\lambda,1)$ under H_1, where $\lambda=\delta/\sqrt{\mathrm{Var}(\hat{\delta})}$ is the noncentrality parameter — how far the distribution shifts when the true effect is δ. Note that the variance is 1 under *both* hypotheses; only the mean differs. We reject H_0 when $|Z|>z_{\alpha/2}$, where $z_{\alpha/2}$ is the upper $\alpha/2$ quantile of the standard normal (e.g., $z_{0.025}=1.96$).

Step 2: the power condition. Power is $P(|Z|>z_{\alpha/2}\mid H_1)$. Under H_1, $Z\sim\mathcal{N}(\lambda,1)$, so

$$\text{power}=P(Z>z_{\alpha/2}\mid H_1)=P\left(\underbrace{Z-\lambda}_{\sim\mathcal{N}(0,1)}>z_{\alpha/2}-\lambda\right)=1-\Phi(z_{\alpha/2}-\lambda).$$

Setting this equal to the desired power $1-\beta$:

$$1-\Phi(z_{\alpha/2}-\lambda)=1-\beta\implies\Phi(z_{\alpha/2}-\lambda)=\beta\implies z_{\alpha/2}-\lambda=-z_\beta,$$

where $z_\beta=\Phi^{-1}(1-\beta)$ (e.g., $z_{0.2}=0.84$ for 80% power). Rearranging gives the key condition:

$$\lambda=z_{\alpha/2}+z_\beta,\quad\text{i.e.,}\quad\frac{\delta}{\sqrt{\mathrm{Var}(\hat{\delta})}}=z_{\alpha/2}+z_\beta.$$

This says: the true signal δ must be large enough relative to the estimation noise $\sqrt{\mathrm{Var}(\hat{\delta})}$ to bridge *both* the false-positive threshold ($z_{\alpha/2}$) and the power gap (z_β) simultaneously.

Step 3: solving for n. Define the total between-person variance in the treatment effect as

$$V_{\text{between}}=(\tau_{1,00}^2+\tau_{0,00}^2)+p^2(\tau_{1,11}^2+\tau_{0,11}^2)+2p(\tau_{1,01}+\tau_{0,01}),$$

so that $\mathrm{Var}(\hat{\delta})=\frac{1}{n}\left[V_{\text{between}}+\sigma^2\left(\frac{1}{T_1}+\frac{1}{T_0}\right)\right]$. Substituting into the power condition and squaring both sides:

$$\frac{\delta^2}{\mathrm{Var}(\hat{\delta})}=(z_{\alpha/2}+z_\beta)^2\implies\frac{n\,\delta^2}{V_{\text{between}}+\sigma^2(1/T_1+1/T_0)}=(z_{\alpha/2}+z_\beta)^2.$$

Dividing numerator and denominator by σ^2 and writing $d=\delta/\sigma$:

$$\frac{n\,d^2}{V_{\text{between}}/\sigma^2+1/T_1+1/T_0}=(z_{\alpha/2}+z_\beta)^2.$$

This single equation — the **joint constraint** — is the core result. It can be rearranged in two useful ways:

Solve for n (given T_0, T_1):

$$\boxed{n \;\geq\; \frac{(z_{\alpha/2}+z_\beta)^2}{d^2}\left[\frac{V_{\text{between}}}{\sigma^2}+\frac{1}{T_1}+\frac{1}{T_0}\right].}$$

Or leave as the joint constraint on (n, T_0, T_1), which makes the trade-offs explicit: the left side is the "budget" your design provides (more participants n and larger effect d increase it), and the right side is the "cost" you must cover (between-person heterogeneity $V_{\text{between}}/\sigma^2$ is a fixed floor that no amount of T can reduce; $1/T_1 + 1/T_0$ is the additional cost from limited time points per arm):

$$\underbrace{\frac{n\,d^2}{(z_{\alpha/2}+z_\beta)^2}}_{\text{design budget}} = \underbrace{\frac{V_{\text{between}}}{\sigma^2}}_{\text{irreducible floor}} + \underbrace{\frac{1}{T_1}+\frac{1}{T_0}}_{\text{cost of short horizon}}.$$

1.5 Numeric Values

For 80% power and $\alpha = 0.05$: $(z_{0.025} + z_{0.2})^2 = (1.96 + 0.84)^2 = 7.84$. Example: moderate heterogeneity ($\tau^2_{a,00} = 0.25$, $\tau^2_{a,11} = 0.25$, $\tau_{a,01} = 0$), $\sigma = 0.5$, $p = 0.2$, equal allocation $T_0 = T_1 = T/2$:

$$V_{\text{between}} = (0.25+0.25) + (0.04)(0.25+0.25) + 0 = 0.52, \qquad \frac{V_{\text{between}}}{\sigma^2} = 2.08.$$

d	Formula (equal allocation)	Example ($p = 0.2$, $T = 50$)
0.2	$n \geq 196\left[V_{\text{between}}/\sigma^2 + 4/T\right]$	$n \geq 196(2.08 + 0.08) = 423$
0.5	$n \geq 31.4\left[V_{\text{between}}/\sigma^2 + 4/T\right]$	$n \geq 31.4(2.08 + 0.08) = 68$

1.6 Caveat for Bandit Settings

The formula assumes the analyst controls T_0 and T_1. In a bandit, arm allocation is adaptive and T_0, T_1 are random. Similarly, $\bar{c}_{i,a}$ may differ between arms if context influences arm selection (which it does, by design). The formula is therefore optimistic, but gives principled (n, T) ranges to avoid degenerate environments.